

The Results section: part 2

Fish 290

Thursday February 6th

Posters are coming up!

- Poster session is Thursday Feb. 20^h (two weeks from today).
- Attendance in class will be mandatory.
- Your poster is worth 20% of your final grade

How can you prepare now?

- Continue re-reading your chosen paper and get to know it well
- Observe the posters from actual studies (see Canvas) and past student posters. What works, what doesn't?

Presentation of Results:

Today's agenda:

1. Review of statistics in excel
2. Reporting results in text
3. Basics of tables

Let's detour for a moment to talk about statistics.

What is the purpose of using a statistical test?

Basics of statistics

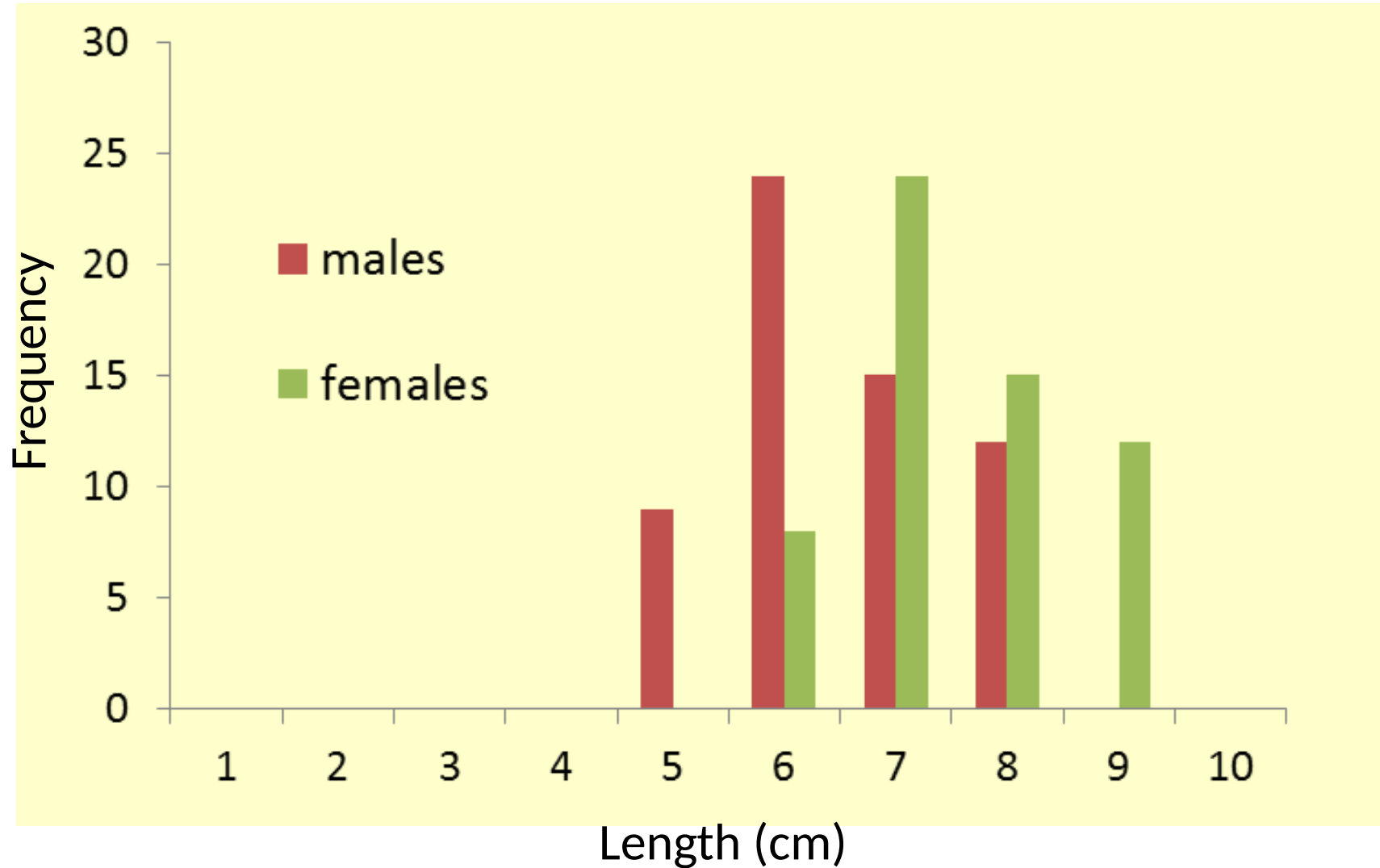
We collect data to test or “falsify” a null hypothesis

Null hypothesis:	Male and female fish are of similar length
Approach:	Measure lengths of males and females
Data:	Length measurements by sex
Attributes:	Mean, standard deviation, sample size
Statistical test:	Two-sample t-test
Output:	Probability of obtaining the data distribution if the null hypothesis were true

Male mean = 6.5, SD = 0.94, n = 60

Female mean = 7.5, SD = 0.94, n = 60

$t = 5.57$, $df = 118$, $P < 0.001$



How do we read this?

Male mean = 6.5 cm, SD = 0.94, n = 60

Female mean = 7.5 cm, SD = 0.94, n = 60

$t = 5.57$, $df = 118$, $P < 0.001$

- **SD** = standard deviation, a measure of variation around the mean; the bigger the number, the more variation
- **n** = sample size
- **t** = the test statistic, this is calculated from the data and compared to a null distribution
- **df** = degrees of freedom (a function of sample size: $(n(\text{males}) - 1) + (n(\text{females}) - 1)$)
- **P** = the probability of obtaining these data if the null hypothesis were true

What does it mean?

Here we can be confident that the lengths of males and females differ. That is, there is a $<0.1\%$ chance that their length distributions were drawn from the same pool.

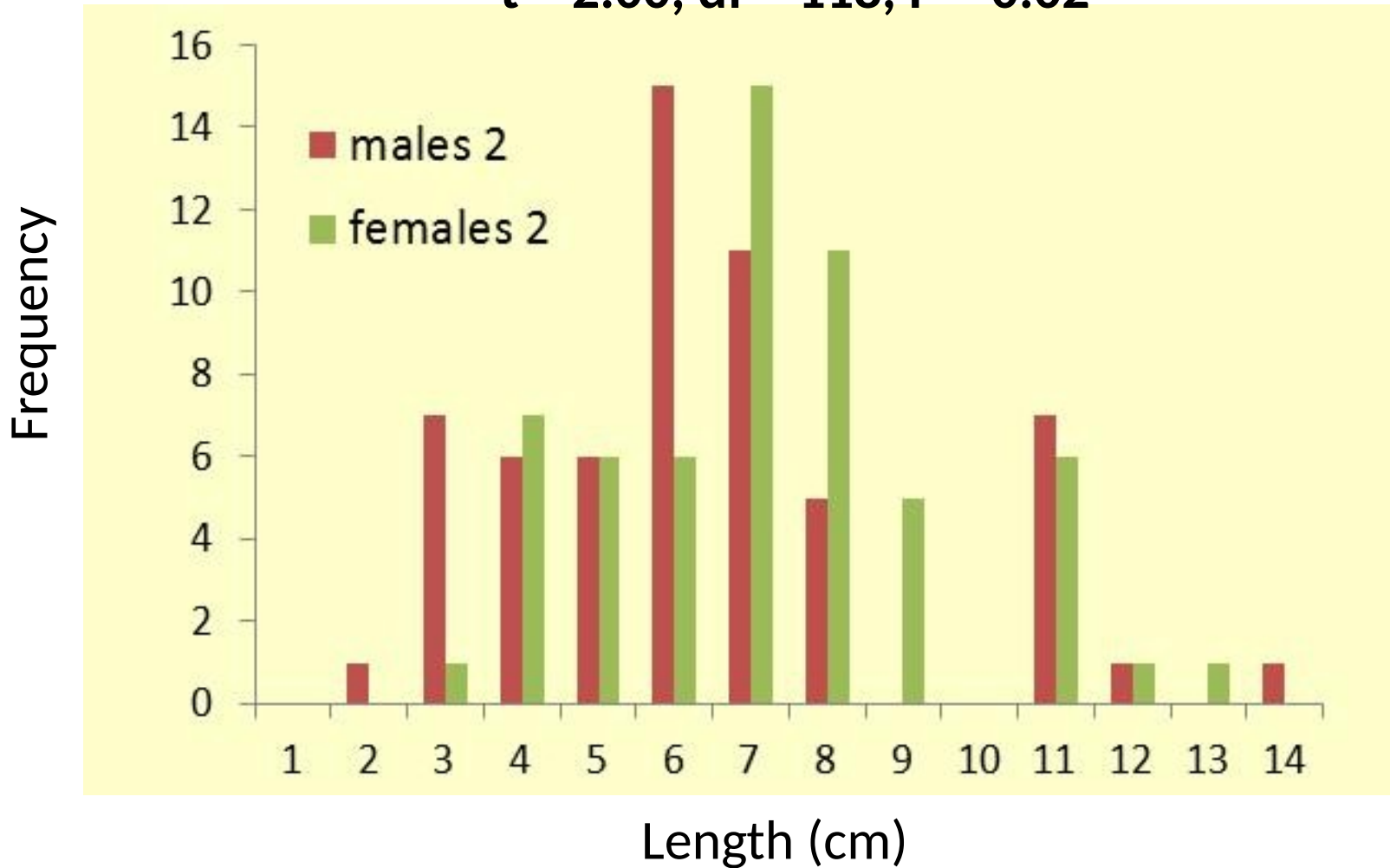
Remember that this assumes that the fish were collected in a random (non-biased manner), and so the sampling methods are important.

How would our confidence change if the mean values were the same, **but the data were more variable?**

Male mean = 6.5, **SD = 2.73**, n = 60

Female mean = 7.5, **SD = 2.73**, n = 60

t = 2.00, df = 118, P = 0.02



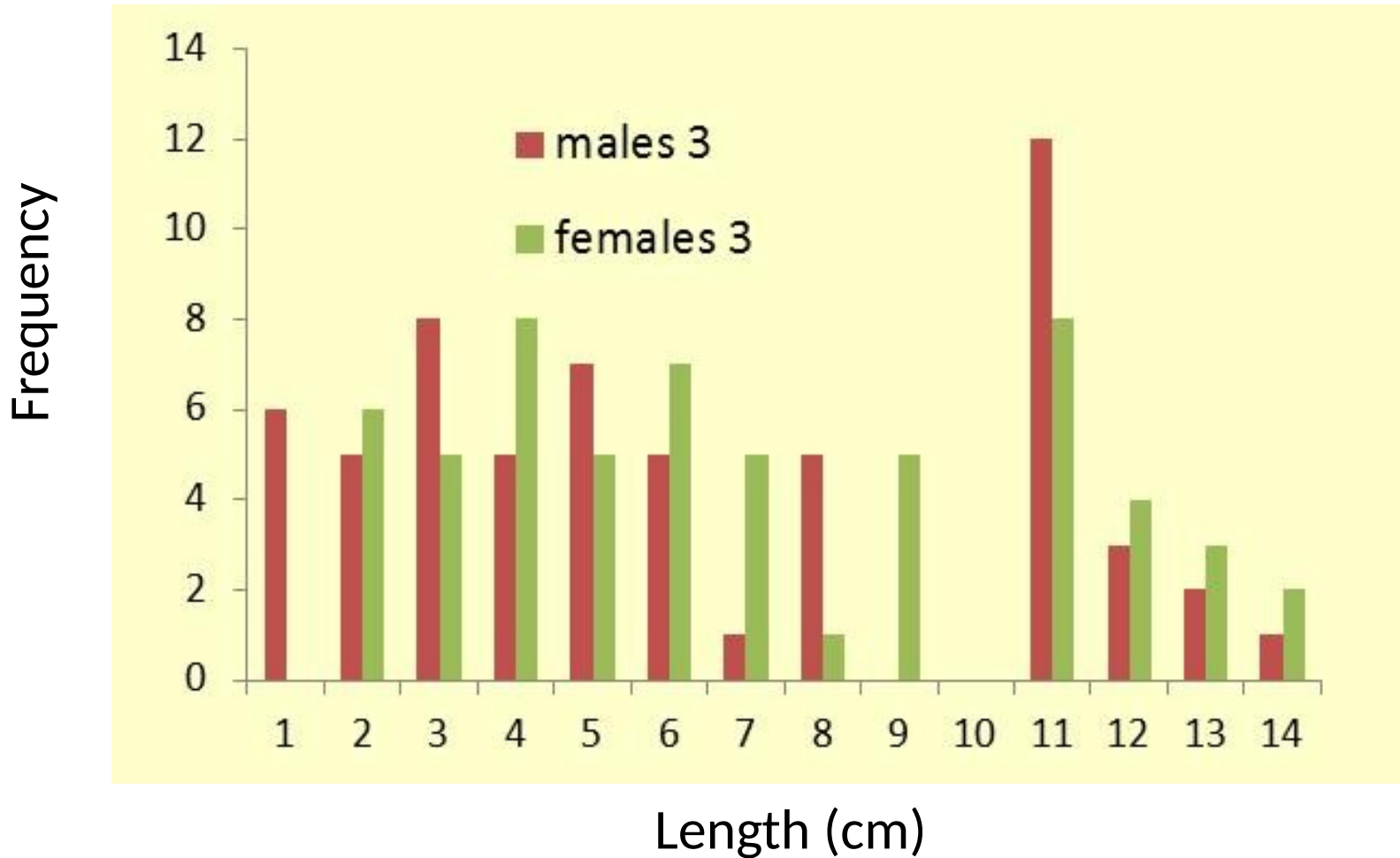
In this case we can still be confident that the lengths of males and females differ. The statistical test indicates that there is a $<2\%$ chance that their distributions were drawn from the same pool.

Suppose **we increase the variation in the data even more** but keep the means the same?

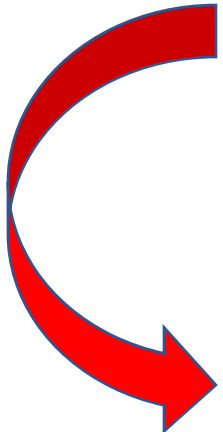
Male mean = 6.5, **SD = 4.19**, n = 60

Female mean = 7.5, **SD = 4.19**, n = 60

t = 1.31, df = 118, P = 0.10



Suppose we take the last data set and
double the sample size.



Male mean = 6.5, SD = 4.19, **n = 60**
Female mean = 7.5, SD = 4.19, **n = 60**
t = 1.31, df = 118, P = 0.10

Male mean = 6.5, SD = 4.19, **n = 120**
Female mean = 7.5, SD = 4.19, **n = 120**
t = 1.86, df = 238, P = 0.03

So, our ability to confidently detect a difference when there is one:

- 1) Sample size: more is better
- 2) Difference between means: bigger is better
- 3) Variation: less is better

Thinking about your results...

Is statistical significance the same thing as biological importance?

NO!!!*

- Can have a sig. difference, but difference is small and/or biologically trivial
- Can have a non-sig. difference with important biological implications

*You have a short reading on this idea this week!

Common types of statistical tests

If you want to...	Use a...
Compare proportions	Chi-squared test
Compare means of two groups	T-test
Compare means of more than two groups	ANOVA
Test dependence of two continuous variables	Linear regression

Lots of other statistical tests exist, but are more complex and beyond the scope of this class

Presentation of Results:

Three complementary approaches

- 1) **Text:** We state in words what we found - what happened.
- 2) **Figures:** Graphs help visualize complex relationships and draw the reader's attention to the most important findings.
- 3) **Tables:** Provide details on the results that are important

(also supplementary material/appendices)

What *is* a result?

1. **Raw data:** data collected from observations or experiments as described in the Methods
2. **Summary statistics:** A number representing a summary of raw data, such as: sample size, mean, median, standard deviation
3. **Patterns:** an increase or decrease, larger or smaller, etc.
4. **Results of statistical tests:** test statistic, p-value, r^2 value, etc.

What is reported in the Results section? Usually the latter three items (why not raw data?)

Try to identify:

1. Raw data
2. Summary statistics
3. Patterns
4. Results of statistical tests

Results

A total of 1068 *A. carbo* (99.6–147.8 cm TL; mean size of 120.0 ± 6.5 cm TL) and 158 *A. intermedius* (104.0–147.5 cm TL; mean size of 127.7 ± 9.3 cm TL) were sampled (Table 1). Significant differences were found in the mean length between spawning ($n = 349$, 120.5 ± 6.7 cm TL) and nonspawning season ($n = 719$, 119.6 ± 6.5 cm TL) for *A. carbo* ($F = 4.273$, $p < 0.05$), with larger individuals being more common during the spawning season, while such differences were not found for *A. intermedius* among seasons (spawning season: $n = 62$, 127.1 ± 10.0 cm TL; nonspawning season: $n = 96$, 128.0 ± 8.9 cm TL; $F = 0.307$, $p > 0.05$; Table 1).

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Presentation of Results:

Compare these sentences describing the same result:

1. The size of seeds is shown in Fig. 1.
2. Fertilized seeds were larger than unfertilized seeds.
3. Fertilized seeds were larger than unfertilized seeds (Fig. 1).
4. The mean weight of fertilized seeds was 1.8 g. The mean weight of unfertilized seeds was 0.9 g.
5. Fertilized seeds were larger than unfertilized seeds (mean 1.8 vs. 0.9 g, $p=0.019$; Fig. 1).

Presentation of Results:

Compare these sentences describing the same result:

1. The size of seeds is shown in Fig. 1.
 - We don't know what the pattern is- text doesn't stand alone!
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 - Asking reader to interpret for you- that is your job!!
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3. Fertilized seeds were larger than unfertilized seeds (Fig. 1).
 - Better, but comes at an opportunity cost- reader needs to find Fig. 1 and break from reading text (e.g. how much larger?)
4. The mean weight of fertilized seeds was 1.8 g. The mean weight of unfertilized seeds was 0.9 g.
 - Asking reader to interpret for you- that is your job!!
5. Fertilized seeds were larger than unfertilized seeds (mean 1.8 vs. 0.9 g, $p=0.019$; Table 3; Fig. 1).
 - Best: gives both the interpretation and the detail needed.

Making effective tables

Tables are appropriate when the precision of a value matters (like a p-value) and you have many variables to display.

Tables are also good for background information (comparing field sites, results from prior literature, or species)

For example:

- A list of species used in study and their locations and diets
- A list of relevant past studies and their key findings
- The results of multiple statistical tests

Making effective tables

We read from left-to-right, so put context-setting information on the left and dependent information on the right.

These tables show the same data. Which is easier to read?

Time (minutes)	Temperature (°C)
0	25
3	27
6	29
9	31
12	32
15	32

Temperature (°C)	Time (minutes)
25	0
27	3
29	6
31	9
32	12
32	15

Making effective tables

Be judicious about significant figures: just because Excel says the mean is 0.4684504, doesn't mean we need that many numbers.

Pay attention to how others report similar types of values.

These tables show the same results. Which is easier to read?

Mean	P-value
45.754	0.131543
31.157	0.032698
131.845	0.146644
14.064	0.0022465
113.254	0.1202454
9.015	0.791545

Mean	P-value
45.7	0.131
31.1	0.032
131.8	0.146
14.0	0.002
113.2	0.120
9.01	0.791

Making effective tables

Some more advice:

- Bold significant p-values so they stand out from non-significant values
- Similar values should match in the # of significant figures (notice the 14.0 and 9.0 values- I added a zero so it matches the others)

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Mean	P-value
45.754	0.131543
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45.7	0.131
31.1	0.032
131.8	0.146
14.0	0.002
113.2	0.120
9.0	0.791

Making effective tables

It is convention to use horizontal borders on the header row and last row, but not all borders.

These tables show the same data. Which is more pleasing to the eye?

Order	Variable A	Variable B	Variable C
Acipenseriformes	27	26	26
Amiiformes	1	1	0
Lepisosteiformes	7	7	5
Albuliformes	11	8	8
Anguilliformes	933	197	191
Elopiformes	9	6	6
Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6

Order	Variable A	Variable B	Variable C
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Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6



Your tables should look like
this

Order	Variable A	Variable B	Variable C
Acipenseriformes	27	26	26
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Elopiformes	9	6	6
Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6

Assignments for today and for home

In class:

1. Continue to make figures and tables for your own data

At home:

2. **By tonight at 11:59 pm upload your peer reviews of Introductions**
3. **By tomorrow at 11:59, upload your Methods (with intro for context), and your most up to date Data**